

ENGELBERG, Switzerland

WMO: 066550

Lat: 46.8217N

Lon: 8.4106E

Elev: 1037

StdP: 89.47

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.2	-11.0	-16.3	1.0	-12.7	-13.7	1.3	-9.9	9.0	7.9	6.8	5.1	1.1	100	0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	26.6	17.8	24.7	17.3	23.0	16.4	18.7	24.7	17.8	23.4	17.0	22.0	3.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.6	13.4	21.4	15.7	12.7	20.4	15.0	12.1	19.5	57.7	24.8	54.6	23.6	51.9	22.1	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	5.4	4.4	DB	-15.5	29.4	2.7	1.4	-17.4	30.4	-19.0	31.2	-20.6	32.0	-22.6	33.1	(4)
(5)			WB	-16.0	20.4	2.7	0.8	-17.9	21.0	-19.5	21.5	-21.0	21.9	-22.9	22.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-1.8	-1.4	2.8	7.2	10.6	14.8	16.1	15.6	11.8	8.4	3.2	-0.6	(6)	
	DBStd	7.64	4.69	4.92	4.52	4.28	3.99	3.91	3.35	3.35	3.49	4.06	4.65	4.60	(7)	
	HDD10.0	1694	366	318	224	103	42	6	1	1	22	76	206	329	(8)	
	HDD18.3	4078	624	552	481	334	242	118	84	98	197	307	454	587	(9)	
	CDD10.0	700	0	0	1	19	60	149	192	174	75	27	3	0	(10)	
	CDD18.3	41	0	0	0	0	1	11	16	12	1	0	0	0	(11)	
	CDH23.3	327	0	0	0	0	9	105	122	90	2	0	0	0	(12)	
	CDH26.7	47	0	0	0	0	1	16	18	13	0	0	0	0	(13)	
(14) Wind	WSAvg	1.6	1.3	1.4	1.7	2.0	1.9	1.7	1.6	1.5	1.4	1.4	1.4	1.3	(14)	
(15) Precipitation	PrecAvg	1646	94	105	109	126	175	183	183	188	136	107	125	116	(15)	
	PrecMax	2069	224	307	216	253	278	283	348	407	233	190	346	289	(16)	
	PrecMin	1252	5	33	43	28	42	109	78	56	54	4	6	5	(17)	
	PrecStd	186	56	62	49	51	70	44	67	73	45	45	68	63	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.4	10.9	15.4	21.0	24.5	28.8	28.7	28.5	23.1	19.9	15.9	13.1	(19)	
		MCWB	6.1	4.6	7.7	11.6	15.3	18.2	18.8	17.9	16.3	11.8	8.3	5.8	(20)	
	2%	DB	8.1	8.4	13.3	18.5	21.8	26.3	26.4	25.8	21.2	18.0	14.2	9.0	(21)	
		MCWB	3.6	3.3	6.5	10.3	13.8	17.5	17.9	17.7	15.2	11.5	7.5	4.3	(22)	
	5%	DB	6.0	6.6	11.5	16.5	19.9	24.2	24.4	23.7	19.3	15.9	11.7	6.4	(23)	
		MCWB	3.1	2.4	5.8	9.2	12.8	16.7	17.4	17.1	14.3	10.9	7.2	3.5	(24)	
	10%	DB	4.2	4.7	9.4	14.5	17.8	22.0	22.6	21.6	17.5	14.0	9.2	4.8	(25)	
		MCWB	2.0	2.0	4.8	8.1	11.7	15.9	16.5	16.4	13.4	10.4	6.4	2.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.7	5.7	8.6	12.2	16.5	19.6	19.9	19.5	17.5	13.8	9.8	6.6	(27)	
		MCDB	10.2	9.0	13.7	19.4	23.0	26.7	26.4	25.2	21.6	16.7	13.0	10.4	(28)	
	2%	WB	4.6	4.3	7.3	10.8	14.8	18.3	18.7	18.5	16.0	12.6	8.5	5.4	(29)	
		MCDB	7.1	7.2	11.9	17.3	20.2	24.5	24.9	24.0	19.8	15.4	12.1	8.4	(30)	
	5%	WB	3.3	3.2	6.2	9.7	13.3	17.2	17.8	17.5	14.9	11.7	7.6	4.1	(31)	
		MCDB	5.3	5.7	10.4	15.4	18.4	23.0	23.4	22.5	18.3	14.6	11.0	6.3	(32)	
	10%	WB	2.2	2.0	5.2	8.7	12.1	16.2	16.9	16.6	13.9	10.8	6.6	2.8	(33)	
		MCDB	4.1	4.5	8.9	13.6	16.8	21.3	21.7	21.0	16.9	13.6	9.4	4.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	6.8	8.3	9.3	9.5	9.6	9.5	8.9	8.1	7.7	6.3	5.8	(35)	
		MCDBR	7.8	8.7	11.0	12.5	13.1	13.3	12.6	12.2	10.6	10.1	9.1	7.6	(36)	
	5% WB	MCWBR	5.1	5.2	6.3	6.7	7.1	7.0	6.4	6.5	6.3	5.6	5.0	4.6	(37)	
		MCWBR	7.2	7.7	10.1	11.9	11.8	12.3	11.8	11.1	9.7	8.6	8.2	7.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.270	0.300	0.350	0.396	0.416	0.421	0.422	0.409	0.376	0.345	0.302	0.267	(40)	
		taud	2.330	2.251	2.179	2.134	2.162	2.203	2.203	2.253	2.327	2.386	2.410	2.358	(41)	
	Ebn at Noon		826	864	865	855	850	846	839	835	831	804	779	786	(42)	
		Edh at Noon	73	98	124	143	144	140	138	126	107	86	69	64	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.45	2.37	3.58	4.42	4.86	5.68	5.60	4.87	3.75	2.45	1.46	1.20	(44)	
	RadStd		0.17	0.29	0.51	0.69	0.48	0.52	0.49	0.46	0.37	0.28	0.20	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (119 neighbors)		N/A	N/A	N/A	+0.72	+0.40	+0.31	N/A	N/A	+58	+16

Nomenclature: See separate page